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Descending aorta and vena cava blood pumps

Abstract

Methods and devices for supporting circulation. The methods may include positioning a blood pump in the arterial vasculature or the venous vasculature. The methods may include positioning a pump portion of the blood pump in a descending aorta, an inferior vena cava, a renal artery, and/or a renal vein. The methods include delivering a pump portion of a blood pump to a target location and rotating one or more impellers to move blood through the pump portion.

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7985442	12/2010	Gong	N/A	N/A
7988728	12/2010	Ayre	N/A	N/A
7993259	12/2010	Kang et al.	N/A	N/A
7993260	12/2010	Bolling	N/A	N/A
7993358	12/2010	O'Brien	N/A	N/A
7998054	12/2010	Bolling	N/A	N/A
7998190	12/2010	Gharib et al.	N/A	N/A
8012079	12/2010	Delgado	N/A	N/A
8012194	12/2010	Edwin et al.	N/A	N/A
8012508	12/2010	Ludwig	N/A	N/A
8029728	12/2010	Lindsay	N/A	N/A
8034098	12/2010	Callas et al.	N/A	N/A
8048442	12/2010	Hossainy et al.	N/A	N/A
8052749	12/2010	Salahieh et al.	N/A	N/A
8070742	12/2010	Woo	N/A	N/A
8070804	12/2010	Hyde et al.	N/A	N/A
8075472	12/2010	Zilbershlag et al.	N/A	N/A
8079948	12/2010	Shifflette	N/A	N/A
8083726	12/2010	Wang	N/A	N/A
8123669	12/2011	Siess et al.	N/A	N/A
8123674	12/2011	Kuyava	N/A	N/A
8133272	12/2011	Hyde	N/A	N/A
RE43299	12/2011	Siess	N/A	N/A
8152035	12/2011	Eart	N/A	N/A
8152845	12/2011	Bourque	N/A	N/A
8153083	12/2011	Briggs	N/A	N/A
8157719	12/2011	Ainsworth et al.	N/A	N/A
8157721	12/2011	Sugiura	N/A	N/A
8157758	12/2011	Pecor et al.	N/A	N/A
8158062	12/2011	Dykes et al.	N/A	N/A
8162021	12/2011	Tomasetti et al.	N/A	N/A
8167589	12/2011	Hidaka et al.	N/A	N/A
8172783	12/2011	Ray	N/A	N/A
8177750	12/2011	Steinbach et al.	N/A	N/A
8187324	12/2011	Webler et al.	N/A	N/A
8197463	12/2011	Intoccia	N/A	N/A
8210829	12/2011	Horvath et al.	N/A	N/A
8241199	12/2011	Maschke	N/A	N/A
8257258	12/2011	Zocchi	N/A	N/A
8257375	12/2011	Maschke	N/A	N/A

8266943	12/2011	Miyakoshi et al.	N/A	N/A
D669585	12/2011	Bourque	N/A	N/A
8277476	12/2011	Taylor et al.	N/A	N/A
8282359	12/2011	Ayre et al.	N/A	N/A
8292908	12/2011	Nieman et al.	N/A	N/A
D671646	12/2011	Bourque et al.	N/A	N/A
8303482	12/2011	Schima et al.	N/A	N/A
8323173	12/2011	Benkowski et al.	N/A	N/A
8323203	12/2011	Thornton	N/A	N/A
8328750	12/2011	Peters et al.	N/A	N/A
8329114	12/2011	Temple	N/A	N/A
8329158	12/2011	Hossainy et al.	N/A	N/A
8366599	12/2012	Tansley et al.	N/A	N/A
8372137	12/2012	Pienknagura	N/A	N/A
8377033	12/2012	Basu et al.	N/A	N/A
8377083	12/2012	Mauch et al.	N/A	N/A
8382695	12/2012	Patel	N/A	N/A
8388565	12/2012	Shifflette	N/A	N/A
8388649	12/2012	Woodard et al.	N/A	N/A
8419609	12/2012	Shambaugh et al.	N/A	N/A
8419944	12/2012	Alkanhal	N/A	N/A
8439909	12/2012	Wang et al.	N/A	N/A
8449444	12/2012	Poirier	N/A	N/A
8454683	12/2012	Rafiee et al.	N/A	N/A
8485961	12/2012	Campbell et al.	N/A	N/A
8496874	12/2012	Gellman et al.	N/A	N/A
8500620	12/2012	Lu et al.	N/A	N/A
8506471	12/2012	Bourque	N/A	N/A
8535211	12/2012	Campbell et al.	N/A	N/A
8535212	12/2012	Robert	N/A	N/A
8538515	12/2012	Atanasoska et al.	N/A	N/A
8545382	12/2012	Suzuki et al.	N/A	N/A
8545447	12/2012	Demarais et al.	N/A	N/A
8562509	12/2012	Bates	N/A	N/A
8568289	12/2012	Mazur	N/A	N/A
8579858	12/2012	Reitan et al.	N/A	N/A
8579967	12/2012	Webler et al.	N/A	N/A
8585572	12/2012	Mehmanesh	N/A	N/A
8586527	12/2012	Singh	N/A	N/A
8591393	12/2012	Walters et al.	N/A	N/A
8591394	12/2012	Peters et al.	N/A	N/A
8591449	12/2012	Hudson	N/A	N/A
8591538	12/2012	Gellman	N/A	N/A
8591539	12/2012	Gellman	N/A	N/A
D696769	12/2012	Schenck et al.	N/A	N/A
8597170	12/2012	Walters et al.	N/A	N/A
8608661	12/2012	Mandrusov et al.	N/A	N/A
8613777	12/2012	Siess et al.	N/A	N/A
8613892	12/2012	Stafford	N/A	N/A
8617239	12/2012	Reitan	N/A	N/A

8631680	12/2013	Fleischli et al.	N/A	N/A
8632449	12/2013	Masuzawa et al.	N/A	N/A
8641594	12/2013	LaRose et al.	N/A	N/A
8657871	12/2013	Limon	N/A	N/A
8657875	12/2013	Kung et al.	N/A	N/A
8668473	12/2013	LaRose et al.	N/A	N/A
8684903	12/2013	Nour	N/A	N/A
8690749	12/2013	Nunez	N/A	N/A
8690823	12/2013	Yribarren et al.	N/A	N/A
8697058	12/2013	Basu et al.	N/A	N/A
8708948	12/2013	Consigny et al.	N/A	N/A
8715151	12/2013	Poirier	N/A	N/A
8715156	12/2013	Jayaraman	N/A	N/A
8715707	12/2013	Hossainy et al.	N/A	N/A
8721516	12/2013	Scheckel	N/A	N/A
8721517	12/2013	Zeng et al.	N/A	N/A
8734331	12/2013	Evans et al.	N/A	N/A
8734508	12/2013	Hastings et al.	N/A	N/A
8739727	12/2013	Austin et al.	N/A	N/A
8740920	12/2013	Goldfarb et al.	N/A	N/A
8741287	12/2013	Brophy et al.	N/A	N/A
8758388	12/2013	Pah	N/A	N/A
8766788	12/2013	D'Ambrosio	N/A	N/A
8777832	12/2013	Wang et al.	N/A	N/A
8790399	12/2013	Frazier et al.	N/A	N/A
8795576	12/2013	Tao et al.	N/A	N/A
8814543	12/2013	Liebing	N/A	N/A
8814776	12/2013	Hastie et al.	N/A	N/A
8814933	12/2013	Siess	N/A	N/A
8815274	12/2013	DesNoyer et al.	N/A	N/A
8821366	12/2013	Farnan et al.	N/A	N/A
8837096	12/2013	Seebruch	N/A	N/A
8840539	12/2013	Zilbershlag	N/A	N/A
8840566	12/2013	Seibel et al.	N/A	N/A
8849398	12/2013	Evans	N/A	N/A
8862232	12/2013	Zarinetchi et al.	N/A	N/A
8864642	12/2013	Scheckel	N/A	N/A
8876685	12/2013	Crosby et al.	N/A	N/A
8882744	12/2013	Dormanen et al.	N/A	N/A
8888675	12/2013	Stankus et al.	N/A	N/A
8894387	12/2013	White	N/A	N/A
8894561	12/2013	Callaway et al.	N/A	N/A
8897873	12/2013	Schima et al.	N/A	N/A
8900060	12/2013	Liebing	N/A	N/A
8905910	12/2013	Reichenbach et al.	N/A	N/A
8927700	12/2014	McCauley et al.	N/A	N/A
8932141	12/2014	Liebing	N/A	N/A
8932197	12/2014	Gregoric et al.	N/A	N/A
8934956	12/2014	Glenn et al.	N/A	N/A
8942828	12/2014	Schecter	N/A	N/A

8944748	12/2014	Liebing	N/A	N/A
8945159	12/2014	Nussbaum	N/A	N/A
8956402	12/2014	Cohn	N/A	N/A
8961387	12/2014	Duncan	N/A	N/A
8961466	12/2014	Steinbach	N/A	N/A
8971980	12/2014	Mace et al.	N/A	N/A
8974519	12/2014	Gennrich et al.	N/A	N/A
8992406	12/2014	Corbett	N/A	N/A
8997349	12/2014	Mori et al.	N/A	N/A
9002468	12/2014	Shea et al.	N/A	N/A
9023010	12/2014	Chiu et al.	N/A	N/A
9028216	12/2014	Schumacher et al.	N/A	N/A
9028392	12/2014	Shifflette	N/A	N/A
9028859	12/2014	Hossainy et al.	N/A	N/A
9033863	12/2014	Jarvik	N/A	N/A
9033909	12/2014	Aihara	N/A	N/A
9039595	12/2014	Ayre et al.	N/A	N/A
9044236	12/2014	Nguyen et al.	N/A	N/A
9056159	12/2014	Medvedev et al.	N/A	N/A
9066992	12/2014	Stankus et al.	N/A	N/A
9067005	12/2014	Ozaki et al.	N/A	N/A
9067006	12/2014	Toellner	N/A	N/A
9072825	12/2014	Pfeffer et al.	N/A	N/A
9078692	12/2014	Shturman et al.	N/A	N/A
9089329	12/2014	Hoarau et al.	N/A	N/A
9089634	12/2014	Schumacher et al.	N/A	N/A
9089635	12/2014	Reichenbach et al.	N/A	N/A
9089670	12/2014	Scheckel	N/A	N/A
9095428	12/2014	Kabir et al.	N/A	N/A
9096703	12/2014	Li et al.	N/A	N/A
9101302	12/2014	Mace et al.	N/A	N/A
9125977	12/2014	Nishimura et al.	N/A	N/A
9127680	12/2014	Yanal et al.	N/A	N/A
9138516	12/2014	Vischer et al.	N/A	N/A
9138518	12/2014	Campbell et al.	N/A	N/A
9144638	12/2014	Zimmermann et al.	N/A	N/A
9162017	12/2014	Evans et al.	N/A	N/A
9168361	12/2014	Ehrenreich et al.	N/A	N/A
9180227	12/2014	Ludwig et al.	N/A	N/A
9180235	12/2014	Forsell	N/A	N/A
9192705	12/2014	Yanai et al.	N/A	N/A
9199020	12/2014	Siess	N/A	N/A
9217442	12/2014	Wiessler et al.	N/A	N/A
D746975	12/2015	Schenck et al.	N/A	N/A
9227002	12/2015	Giridharan et al.	N/A	N/A
9239049	12/2015	Jarnagin et al.	N/A	N/A
9265870	12/2015	Reichenbach et al.	N/A	N/A
9278189	12/2015	Corbett	N/A	N/A
9283314	12/2015	Prasad et al.	N/A	N/A
9291591	12/2015	Simmons et al.	N/A	N/A

9295550	12/2015	Nguyen et al.	N/A	N/A
9295767	12/2015	Schmid et al.	N/A	N/A
9308302	12/2015	Zeng	N/A	N/A
9308304	12/2015	Peters et al.	N/A	N/A
9314558	12/2015	Er	N/A	N/A
9314559	12/2015	Smith et al.	N/A	N/A
9328741	12/2015	Liebing	N/A	N/A
9333284	12/2015	Thompson et al.	N/A	N/A
9339596	12/2015	Roehn	N/A	N/A
9345824	12/2015	Mohl et al.	N/A	N/A
9358329	12/2015	Fitzgerald et al.	N/A	N/A
9358330	12/2015	Schumacher	N/A	N/A
9364255	12/2015	Weber	N/A	N/A
9364592	12/2015	McBride et al.	N/A	N/A
9370613	12/2015	Hsu et al.	N/A	N/A
9375445	12/2015	Hossainy et al.	N/A	N/A
9381285	12/2015	Ozaki et al.	N/A	N/A
9387284	12/2015	Heilman et al.	N/A	N/A
9409012	12/2015	Eidenschink et al.	N/A	N/A
9416783	12/2015	Schumacher et al.	N/A	N/A
9416791	12/2015	Toellner	N/A	N/A
9421311	12/2015	Tanner et al.	N/A	N/A
9433713	12/2015	Corbett et al.	N/A	N/A
9435450	12/2015	Muennich	N/A	N/A
9446179	12/2015	Keenan et al.	N/A	N/A
9452249	12/2015	Kearsley et al.	N/A	N/A
9474840	12/2015	Siess	N/A	N/A
9486565	12/2015	Göllner et al.	N/A	N/A
9492601	12/2015	Casas et al.	N/A	N/A
9504491	12/2015	Callas et al.	N/A	N/A
9511179	12/2015	Casas et al.	N/A	N/A
9512839	12/2015	Liebing	N/A	N/A
9522257	12/2015	Webler	N/A	N/A
9526818	12/2015	Kearsley et al.	N/A	N/A
9533084	12/2016	Siess et al.	N/A	N/A
9533085	12/2016	Hanna	N/A	N/A
9539378	12/2016	Tuseth	N/A	N/A
9550017	12/2016	Spanier et al.	N/A	N/A
9555173	12/2016	Spanier	N/A	N/A
9555175	12/2016	Bulent et al.	N/A	N/A
9555177	12/2016	Curtis et al.	N/A	N/A
9556873	12/2016	Yanal et al.	N/A	N/A
9561309	12/2016	Glauser et al.	N/A	N/A
9561313	12/2016	Taskin	N/A	N/A
9572915	12/2016	Heuring	N/A	A61M 60/33
9592328	12/2016	Jeevanandam et al.	N/A	N/A
9603983	12/2016	Roehn et al.	N/A	N/A
9603984	12/2016	Romero et al.	N/A	N/A
9611743	12/2016	Toeliner et al.	N/A	N/A

9612182	12/2016	Olde et al.	N/A	N/A
9616157	12/2016	Akdis	N/A	N/A
9616159	12/2016	Anderson et al.	N/A	N/A
9623163	12/2016	Fischi	N/A	N/A
9631754	12/2016	Richardson et al.	N/A	N/A
9642984	12/2016	Schumacher et al.	N/A	N/A
9656010	12/2016	Burke	N/A	N/A
9656030	12/2016	Webler et al.	N/A	N/A
9662211	12/2016	Hodson et al.	N/A	N/A
9669141	12/2016	Parker et al.	N/A	N/A
9669142	12/2016	Spanier et al.	N/A	N/A
9669143	12/2016	Guerrero	N/A	N/A
9675450	12/2016	Straka et al.	N/A	N/A
9675738	12/2016	Tanner et al.	N/A	N/A
9675739	12/2016	Tanner et al.	N/A	N/A
9675742	12/2016	Casas et al.	N/A	N/A
9687596	12/2016	Poirier	N/A	N/A
9687630	12/2016	Basu et al.	N/A	N/A
9700659	12/2016	Kantrowitz et al.	N/A	N/A
9713662	12/2016	Rosenberg et al.	N/A	N/A
9713663	12/2016	Medvedev et al.	N/A	N/A
9715839	12/2016	Pybus et al.	N/A	N/A
9717615	12/2016	Grandt	N/A	N/A
9717832	12/2016	Taskin et al.	N/A	N/A
9717839	12/2016	Hashimoto	N/A	N/A
9726195	12/2016	Cecere et al.	N/A	N/A
9731058	12/2016	Siebenhaar et al.	N/A	N/A
9731101	12/2016	Bertrand et al.	N/A	N/A
9737361	12/2016	Magana et al.	N/A	N/A
9737651	12/2016	Wampler	N/A	N/A
9744280	12/2016	Schade et al.	N/A	N/A
9744287	12/2016	Bulent et al.	N/A	N/A
9750859	12/2016	Bulent et al.	N/A	N/A
9757502	12/2016	Burke et al.	N/A	N/A
9770202	12/2016	Ralston et al.	N/A	N/A
9770543	12/2016	Tanner et al.	N/A	N/A
9771801	12/2016	Schumacher et al.	N/A	N/A
9775930	12/2016	Michal et al.	N/A	N/A
9782279	12/2016	Kassab	N/A	N/A
9782527	12/2016	Thomas et al.	N/A	N/A
9795780	12/2016	Serna et al.	N/A	N/A
9801987	12/2016	Faman et al.	N/A	N/A
9801992	12/2016	Giordano et al.	N/A	N/A
9821098	12/2016	Horvath et al.	N/A	N/A
9821146	12/2016	Tao et al.	N/A	N/A
9827356	12/2016	Muller et al.	N/A	N/A
9833314	12/2016	Corbett	N/A	N/A
9833550	12/2016	Siess	N/A	N/A
9833551	12/2016	Criscione et al.	N/A	N/A
9839734	12/2016	Menon et al.	N/A	N/A

9844618	12/2016	Muller-Spanka et al.	N/A	N/A
9850906	12/2016	Ozaki et al.	N/A	N/A
9855437	12/2017	Nguyen et al.	N/A	N/A
9861504	12/2017	Abunassar et al.	N/A	N/A
9861731	12/2017	Tamburino	N/A	N/A
9872948	12/2017	Siess	N/A	N/A
9878087	12/2017	Richardson et al.	N/A	N/A
9878169	12/2017	Hossainy	N/A	N/A
9889242	12/2017	Pfeffer et al.	N/A	N/A
9895244	12/2017	Papp et al.	N/A	N/A
9895475	12/2017	Toellner et al.	N/A	N/A
9907890	12/2017	Muller	N/A	N/A
9907892	12/2017	Broen et al.	N/A	N/A
9913937	12/2017	Schwammenthal et al.	N/A	N/A
9918822	12/2017	Abunassar et al.	N/A	N/A
9919085	12/2017	Throckmorton et al.	N/A	N/A
9919088	12/2017	Bonde et al.	N/A	N/A
9919089	12/2017	Garrigue	N/A	N/A
9950101	12/2017	Smith et al.	N/A	N/A
9956410	12/2017	Deem et al.	N/A	N/A
9962258	12/2017	Seguin et al.	N/A	N/A
9974893	12/2017	Toellner	N/A	N/A
9974894	12/2017	Morello	N/A	N/A
9981078	12/2017	Jin et al.	N/A	N/A
9985374	12/2017	Hodges	N/A	N/A
9987407	12/2017	Grant et al.	N/A	N/A
10010273	12/2017	Sloan et al.	N/A	N/A
10022499	12/2017	Galasso	N/A	N/A
10028835	12/2017	Kermode et al.	N/A	N/A
10029037	12/2017	Muller et al.	N/A	N/A
10029038	12/2017	Hodges	N/A	N/A
10029039	12/2017	Dague et al.	N/A	N/A
10031124	12/2017	Galasso	N/A	N/A
10034972	12/2017	Wampler et al.	N/A	N/A
10039873	12/2017	Siegenthaler	N/A	N/A
10046146	12/2017	Manderfeld et al.	N/A	N/A
10052419	12/2017	Er	N/A	N/A
10058349	12/2017	Gunderson et al.	N/A	N/A
10058641	12/2017	Mollison et al.	N/A	N/A
10058652	12/2017	Tsoukalis	N/A	N/A
10058653	12/2017	Wang et al.	N/A	N/A
10077777	12/2017	Horvath et al.	N/A	N/A
10080828	12/2017	Wiesener et al.	N/A	N/A
10080834	12/2017	Federspiel et al.	N/A	N/A
10080871	12/2017	Schumacher et al.	N/A	N/A
10208763	12/2018	Schumacher et al.	N/A	N/A
10357598	12/2018	Aboul-Hosn et al.	N/A	N/A
10569005	12/2019	Solem et al.	N/A	N/A
10722631	12/2019	Salahieh et al.	N/A	N/A

10881770	12/2020	Tuval et al.	N/A	N/A
10894115	12/2020	Pfeffer et al.	N/A	N/A
11033727	12/2020	Tuval	N/A	A61M 60/416
11123538	12/2020	Epple et al.	N/A	N/A
11185677	12/2020	Salahieh et al.	N/A	N/A
11229784	12/2021	Salahieh et al.	N/A	N/A
11268521	12/2021	Toellner	N/A	N/A
11280345	12/2021	Bredenbreuker et al.	N/A	N/A
11850413	12/2022	Zeng et al.	N/A	N/A
12017056	12/2023	Guo et al.	N/A	N/A
2001/0003802	12/2000	Vitale	N/A	N/A
2001/0023369	12/2000	Chobotov	N/A	N/A
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2002/0068848	12/2001	Zadini et al.	N/A	N/A
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2002/0128709	12/2001	Pless	N/A	N/A
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2006/0129082	12/2005	Rozga	N/A	N/A
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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of priority of U.S. Provisional App. No. 62/946,927, filed Dec. 11, 2019, and U.S. Provisional App. No. 62/951,519, filed Dec. 20, 2019, the complete disclosures of which are incorporated by reference herein for all purposes.

INCORPORATION BY REFERENCE
(1) All publications and patent applications mentioned in this specification are herein incorporated by reference to the same extent as if each individual publication or patent application was specifically and individually indicated to be incorporated by reference.
(2) The following publications are incorporated by reference herein for all purposes: WO2018/226991A1, WO2019/094963A1, WO2019/152875A1, WO2020/028537A1, WO2020/073047A1, WO2018/226991A1, WO2019/094963A1, WO2019/152875A1, U.S. Pat. No. 9,572,915, and US 2017/0100527.
BACKGROUND
(3) Descriptions have been presented that attempt to increase renal artery and kidney perfusion using a blood pump positioned in a descending aorta. One or more aspects of those descriptions, however, may have deficiencies that can be addressed by catheter-based blood pumps and methods of placement and use that are set forth herein.
(4) Additionally, descriptions have been presented that attempt to reduce blood pressure within one or more renal vein by placing a blood pump in a vena cava and pumping blood away from the region. One or more aspects of those descriptions, however, may have deficiencies that can be addressed by catheter-based blood pumps and methods of placement and use that are set forth herein.
SUMMARY OF THE DISCLOSURE
(5) One aspect of the disclosure is a method of supporting circulation in a descending aorta of a subject. The method may include advancing an expandable pump portion of a blood pump into a descending aorta of a subject, the pump portion comprising an expandable blood conduit between a pump inflow and a pump outflow, the pump portion further including first and second expandable impellers at least partially within the blood conduit. The method may further include expanding the expandable blood conduit to an expanded and deployed configuration within the descending aorta. The method may further include expanding the first expandable impeller into an expanded configuration at least partially within the blood conduit and expanding the second expandable impeller into an expanded configuration at least partially within the blood conduit, and rotating the

first and second expandable impellers to thereby move blood into the pump, through the blood conduit, and out of the pump. The rotating step may move blood through the blood conduit at a rate of at least 3.5 L/min.
(6) In this aspect, the rotating step may cause the pump outflow to include a radial flow component.
(7) In this aspect, moving blood out of the pump outflow may perfuse at least one renal artery.
(8) In this aspect, expanding the expandable blood conduit may comprise expanding the expandable blood conduit to a deployed configuration within the descending aorta such that the pump outflow is upstream or aligned with a renal artery, and wherein the outflow is at least partially directed radially into the renal artery due to the position of the blood conduit in the descending aorta.
(9) In this aspect, expanding the expandable blood conduit to a deployed configuration within the descending aorta may comprise expanding the expandable blood conduit to a deployed configuration near a renal artery such that the pump outflow perfuses at least one renal artery.
(10) In this aspect, expanding the second expandable impeller may comprise expanding the second expandable impeller into an expanded configuration such that at least a portion of the second expandable impeller is extending beyond a proximal end of the blood conduit, and wherein the outflow may have a radial flow component due at least partially to the portion of the second expandable impeller that is extending beyond a proximal end of the blood conduit.
(11) In this aspect, expanding the expandable blood conduit within the descending aorta may comprise expanding a distal scaffold to an expandable configuration that provides radial support to the blood conduit at the location of the first impeller. Expanding the expandable blood conduit within the descending aorta may comprise expanding a proximal scaffold to an expandable configuration that provides radial support to the blood conduit at the location of the second impeller.
(12) In this aspect, expanding the expandable blood conduit may comprise expanding a central region of the blood conduit that is between the first and second impellers, wherein the central region may be more flexible than regions of the blood conduit that surround the first and second impellers.
(13) In this aspect, expanding the expandable blood conduit to a deployed configuration within the descending aorta may comprise expanding the expandable blood conduit to a deployed configuration within the descending thoracic aorta.
(14) One aspect of this disclosure is a method of supporting circulation in a descending aorta of a subject. The method may include advancing an expandable pump portion of a blood pump into a descending aorta of a subject, the pump portion including a distal impeller and a proximal impeller; expanding a distal expandable blood conduit into an expanded configuration aligned with or upstream to a renal artery; expanding the distal impeller to an expanded configuration at least partially within the distal expandable blood conduit; expanding a proximal expandable blood conduit into an expanded configuration downstream from the renal artery; expanding the proximal impeller to an expanded configuration at least partially within the proximal expandable blood conduit; rotating the distal impeller to move blood into a distal end of the distal expandable blood conduit, through the distal expandable blood conduit, and out of a proximal end of the distal expandable blood conduit; and rotating the proximal impeller to move blood through the proximal expandable blood conduit.
(15) In this aspect, rotating the distal impeller may cause blood to move out of the proximal end of the distal expandable blood conduit and perfuse the renal artery, optionally also perfusing a second renal artery.
(16) In this aspect, rotating distal and proximal impellers may move blood past the distal and proximal impellers in an antegrade direction.
(17) In this aspect, rotating the distal impeller may move blood past the distal impeller in an antegrade direction, and rotating the proximal impeller may move blood past the proximal impeller

in a retrograde direction toward the renal artery.
(18) In this aspect, rotating the distal and proximal impellers may be performed discontinuously and in a manner that is related to one or more aspects of the subject's cardiac cycle.
(19) In this aspect, rotating the distal and proximal impellers may occur during at least a portion of systole, and wherein the distal and proximal impellers may not be rotated during at least a portion of diastole, optionally during any of diastole.
(20) In this aspect, rotating the proximal impeller may move blood into a distal end of the proximal expandable blood conduit, through the proximal expandable blood conduit, and out of a proximal end of the proximal expandable blood conduit.
(21) In this aspect, expanding a distal expandable blood conduit into an expanded configuration aligned with or upstream to a renal artery may comprise expanding a distal expandable scaffold.
(22) In this aspect, expanding a proximal expandable blood conduit may comprise expanding a proximal expandable scaffold.
(23) In this aspect, rotating the distal and proximal impellers may be performed by rotating a common drive mechanism to which the distal and proximal impellers are in rotational communication.
(24) In this aspect, the method may further comprise causing blood to flow radially outward from a proximal end of the distal blood conduit.
(25) One aspect of the disclosure is a method of supporting circulation in proximity to renal veins. The method may include advancing an expandable pump portion of a blood pump into an inferior vena cava ("IVC") of a subject, the pump portion comprising an expandable blood conduit between a proximal end and a distal end, the pump portion further including first and second expandable impellers; expanding the expandable blood conduit to a deployed configuration within the IVC; expanding the first expandable impeller into an expanded configuration at least partially within the blood conduit; expanding the second expandable impeller into an expanded configuration at least partially within the blood conduit; and rotating the first and second expandable impellers to move blood into the blood conduit, through the blood conduit, and out of the blood conduit. This aspect may additionally include any other suitable method step described herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is an example of an expandable pump portion secured in a descending aorta.
(2) FIG. **2** is an example of an expandable pump portion with one or more inflatable anchors secured in a descending aorta.
(3) FIG. **3** is an example of an expandable pump portion secured in a descending aorta.
(4) FIG. **4** is an example of an expandable pump portion secured in a descending aorta.
(5) FIG. **5** is an example of an expandable pump portion secured in an inferior vena cava.
DETAILED DESCRIPTION
(6) The disclosure is related to catheter blood pumps that may be placed in one or more of the following locations: a descending aorta, an inferior vena cava ("IVC"), a renal artery, or a renal vein. The blood pumps herein may include pump portions that include one or more impellers within a blood conduit, wherein the one or more impellers are sized and configured to move blood through the blood conduit when rotated.
(7) One aspect of the disclosure is related to intravascular multiple impeller blood pumps that include pump portions that are sized and configured for placement in the descending aorta, optionally within the abdominal aorta, such as is shown in the examples in FIGS. **1-3**. The intravascular multiple impeller blood pumps may be configured for cardiovascular and circulatory support. The entirety of U.S. Pat. No. 9,572,915, including the background section thereof, is fully

incorporated by reference herein in this context. For example, intravascular multiple impeller blood pumps herein may be used in cardiovascular support, and may be used to facilitate perfusion of targeted vessel and/or organs, such as the kidneys.
(8) One aspect of the disclosure is related to intravascular multiple impeller blood pumps that include pump portions that are sized and configured for placement in an inferior vena cava ("IVC"). The background section of US 2017/0100527 is fully incorporated by reference herein in this context. For example, intravascular multiple impeller blood pumps herein may be placed in the vicinity of a junction of an IVC and one or more renal veins, and may be used to facilitate blood flow in the venous vasculature, optionally to reduce renal venous pressure. Any of the disclosure in US 2017/0100527 related to delivery of a blood pump to a target location near the renal veins is incorporated by reference herein for all purposes, including any methods herein that include delivering a blood pump to a target location and/or expanding a blood pump at a target location. The disclosure in WO2018/226991A1, WO2019/094963 A1, WO2019/152875A1, WO2020/028537A1, WO2020/073047A1 may include exemplary multiple impeller blood pumps, any of which may be delivered, deployed, and operated to move fluid therethrough for cardiovascular and circulatory support according to any of the methods herein. In any of the relevant methods herein, a multiple impeller blood pump may be delivered to a location in a patient's descending aorta, such as in the abdominal aorta.
(9) FIGS. **1-3** illustrate exemplary pump portions of multiple impeller blood pumps deployed in expanded configurations in the descending aorta, with the impellers shown expanded and rotating to pump blood through the blood conduit of the pump portion of the blood pump. The deployed and expanded pump portions shown in FIGS. **1-3** may be similar to the pump portions shown and described in WO2018/226991A1, WO2019/094963 A1, WO2019/152875A1, WO2020/028537A1, WO2020/073047A1, and any suitable disclosure in any of these publications incorporated fully by reference herein for all purposes may be incorporated by reference into the description of FIGS. **1-3** herein.
(10) An exemplary advantage of the blood pumps and methods shown in FIGS. **1-3** is that the multiple impeller pump may be able to achieve higher flow rates than some existing single impeller blood pumps that are positioned in the descending aorta. The advantages of relatively low-profile and higher flow rate expandable blood pumps may be described in more detail in WO2018/226991A1, WO2019/094963 A1, WO2019/152875A1, WO2020/028537A1, WO2020/073047A1, and similarly apply to the relatively low-profile and higher flow rate blood pumps positioned for use in the descending aorta, such as those shown in FIGS. **1-3**, or for use in a vena cava, such as shown in FIG. **5**. For example, without limitation, the pump portions of the blood pumps herein may be sized and configured such that they can be introduced via a **10**F introducer, while being able to achieve flow rates of at least 3.5 L/min when expanded and activated, which may be higher than some existing pump portions that are designed for placement in the descending aorta or vena cava.
(11) Any of the blood pumps herein may include one or more anchors (which may also be referred to herein as stabilizing members) that are sized and configured to help stabilize a pump portion of the blood pump with respect to the descending aorta or IVC, examples of which are shown in FIGS. **1** and **2**. To anchor or stabilize as used in this context refers generally to being anchored in the aorta (or IVC) to prevent axial migration of the pump relative to the aorta, although some minimal movement may still occur, such as due to slight deformation of one or more anchors when the pump is in use. FIG. **1** illustrates an exemplary pump portion **100** of a blood pump in an expanded configuration stabilized in place in descending aorta **150** and generally proximate to renal arteries **152**, with most or all of pump portion **100** upstream the renal arteries. Pump portion **100** includes one or more anchors or stabilizing members (**120**, **122**, **124**, **126**) that in their expanded configurations extends radially outward and optionally also distally from an expandable blood conduit or shroud **102** of the blood pump, as shown. Pump portion **100** may include one or

more anchors, and any of those shown in FIG. **1** may be optional. For example, the pump may only have one or more distal stabilizing members **120** extending from a distal end or distal region of the expandable blood conduit, as shown. It may be desirable to have anchors at one or more ends or end regions of the pump, such as anchors **120** and anchors **126**, and/or stabilizing members that are disposed axially between the fluid conduit ends, such as one or more of stabilizing members **122** or stabilizing members **124**.

(12) Any of the anchors or stabilizing members herein may comprise one or more materials such that the anchor is adapted to self-expand to an at least partially expanded configuration. For example, any of the anchors or stabilizing members herein may be adapted to self-expand after a sheath, and example of which is shown in FIG. **1** as sheath **130**, is retracted proximally to allow the anchor(s) to expand. Any of the anchors herein may be re-sheathed before or after final pump deployment and activation if pump repositioning is needed, or if the pump is to be re-sheathed and removed from the patient, such as with sheath **130** shown in FIG. **1**. Sheath **130** is adapted to be axially movable relative to the catheter and pump portions that are shown to deploy and sheath the expandable pump portion, which may be described in more detail in WO2018/226991A1, WO2019/094963 A1, WO2019/152875A1, WO2020/028537A1, WO2020/073047A1. Any of the anchors herein may comprises a deformable material, such as nitinol, and may be coupled to a scaffold section of the expandable pump portion. Coupled to in this context includes being integral with (e.g., unitary with) and being indirectly attached.

(13) Anchors herein may have elongate finger or arm-like configurations, extending radially outward when deployed, such as exemplary anchors **120**, **122**, **124** and **126** shown in FIG. **1**. Configurations in which they extend distally and radially relative to where they are coupled to the blood conduit, examples of which are shown in FIG. **1**, may ease or facilitate the re-collapse and capture of the anchors when a sheath is advanced distally (relative motion) over the blood conduit and anchors.

(14) The entirety of any of the anchors herein need not extend distally relative to the blood conduit to be able to facilitate collapse. For example, stabilizing members **124** may be considered to have a general curved, C, U, or bow shape when expanded, and initially extend radially and distally relative to the blood conduit, then extend further proximally at their radially outer ends as shown. When the sheath is advanced distally, stabilizing members **124** are adapted to be collapse and be re-sheathed. Additionally, any of the anchors herein may extend solely radially outward when expanded and will be able to be collapsed.

(15) Any of the anchors herein may be separate components from the blood conduit (not unitary therewith), but may be secured to one or more other blood conduit components during manufacture. For example, any of the anchors herein may be secured to any portion of an expandable scaffold, examples of which are described in examples incorporated by reference herein. Anchors may optionally and alternatively be secured to and extend from a distal strut or a proximal strut of the expandable pump portion (not shown).

(16) Anchors herein may be unitarily formed with one or more blood conduit components. For example, any of the anchors and scaffolds herein may be laser cut from the same nitinol tubular starting material. Forming the anchors unitarily with one or more blood conduit components may simplify construction by eliminating a step of coupling an anchor to a region of the blood conduit.

(17) Any of the anchors herein may have distal free ends that are configured to further stabilize the anchor against tissue. For example, without limitation, any of the anchors herein may include an end that include one or more of barbs, protrusions, or any other type of element or configuration that is configured to increase the stability of the anchor with respect to the vessel wall, such as a descending aorta wall or IVC.

(18) Any of the anchors herein may be equidistantly spaced apart around the shroud (circumferentially). For example, first and second anchors may be circumferentially spaced 180 degrees from each other, three anchors may be spaced 120 degrees from each other, four anchors

may be spaced 90 degrees from each other, etc. In other embodiments, any of the anchors may not be circumferentially spaced equidistantly around the blood conduit.
(19) FIG. **1** also shows exemplary distal expandable and collapsible struts **110**, exemplary proximal expandable and collapsible struts **108**, exemplary distal impeller **104**, exemplary proximal impeller **106**
(20) Any of anchors herein may optionally be adapted to be inflatable, such as anchor **260′** and **260″** shown in the embodiment in FIG. **2**. Any aspect of any other pump portion herein may be incorporated into the expandable pump portion shown in FIG. **2** (and vice versa), such as an expandable blood conduit, scaffold, membrane, impeller(s), and struts, examples of which are shown. Inflatable anchors **260′** and **260″** may be in fluid communication with inflation fluid in an inflation fluid source (not shown for clarity), which may be external to the patient. The pump portion in this embodiment has an inflatable anchor that comprises a distal inflatable anchor region **260″** and a proximal inflatable anchor region **260′**, which are in fluid communication in this example with each other via an inflation line **270**. Inflation lumen **260** is also in fluid communication with anchor region **260′** and extends along one of the proximal struts and then proximally through the catheter portion of the blood pump, as shown in FIG. **2**.
(21) Any of the blood pumps herein may include an inflation lumen that extends to or towards an external portion of the blood pump, such as an external console or other external control portion of the pump. The inflation lumen may be in communication with an inflation fluid source, such as a gas source or a liquid source. One or more external pumps may help control the inflation of inflation fluid to the inflatable anchor(s). Inflation of the one or more inflatable anchors may occur at any time after the pump portion is deployed from a delivery sheath, such as sheath **220** as shown in FIG. **2**. Any the inflatable anchors herein may be coupled to the expandable blood conduit using a wide variety to bonding concepts, such as spot welding or other techniques to couple inflatable members to an expandable blood conduit.
(22) Any of the inflatable anchors herein may be configured to allow for some blood flow around the pump portion, before, during, and/or after operation of the pump. For example, any inflatable anchor herein may have an at least partially helical configuration when inflated, such as the anchor regions **260′** and **260″** shown in FIG. **2**, which can allow blood to continue to flow around the expandable blood conduit without the inflatable element completely occluding blood flow in the descending aorta or IVC. FIG. **2** also shows renal arteries **252**.
(23) Any of the inflatable anchors herein may be sized and configured such that when inflated and expanded with a fluid, they expand and engage the descending aorta wall **250** and help anchor the pump in place to minimize axial migration of the blood conduit during use.
(24) The outer dimension of any of the pump portions herein may be varied as needed for the implantation and/or the placement location within the patient. For example, the pump portion may be sized such that in an expanded configuration, an outer surface (optionally cylindrical) of the blood conduit directly engages or substantially engages a descending aorta or IVC sufficiently to anchor the pump in place and sufficiently minimize movement. An exemplary pump that is sized and configured to be expanded and anchored against a descending aorta **350** is shown in FIG. **3**. The advantages of a multiple impeller pump apply in this embodiment, even if in this example the expandable pump portion is not collapsible to French sizes comparable to pump designs that have one or more anchors, for example. Any of the disclosure herein related to multi-impeller pumps is incorporated by reference into FIG. **3** and may be integrated into the example show in FIG. **3**. FIG. **3** also illustrates exemplary distal impeller **304**, proximal impeller **306**, renal arteries **352** being perfused by the activated pump portion, and sheath **330** that can be used to deliver the expandable pump portion.
(25) Exemplary descending aorta positions or locations for any of the pump portions herein are shown in FIGS. **1-3**, with the pump portions positioned just upstream and proximate the ostia of the renal arteries, as shown. The position of the pump portions may be such that the proximal end of

the blood conduit is aligned with, substantially aligned with, or just upstream, one or both of the ostia of the renal arteries. The outflow of the pump portions, as shown in FIGS. **1-3**, may include a radial component. Positioning the pump just upstream to or aligned with one or both renal arteries may help direct blood towards or into the renal arteries, which may help perfuse the renal arteries and kidneys.

(26) In alternative embodiments not shown in FIGS. **1-3**, the multiple impeller pump portion is not expandable and collapsible. Non-expandable pumps may be positioned in the descending aorta as shown in any of FIGS. **1-3** just upstream to or aligned with a renal artery such that the pump outflow is still at least partially directed towards one or both renal arteries.

(27) FIG. **4** illustrates an exemplary embodiment of a pump portion that is similar to the pump portions shown in FIG. **1-3**, and applicable disclosure related to FIGS. **1-3** is incorporated by reference into the embodiment of FIG. **4**. The blood pump shown expanded in place in FIG. **4** does not have a fluid conduit that extends all the way in between distal and proximal impellers, as shown. The pump portion in FIG. **4** includes a distal pump portion **480** that includes an expandable scaffold or basket, and a proximal pump portion **490** that includes an expandable scaffold or basket, each of which at least partially surrounds an impeller (distal impeller **482** is labeled). In this embodiment, the impellers may be rotated by the same or an otherwise common drive mechanism **472** (e.g., a drive cable or drive shaft), which may be rotated by an external motor that is also in rotational communication with the impellers. The distal pump portion **480** includes a distal blood conduit with an inflow and an outflow as shown, and the proximal pump portion **490** includes a proximal blood conduit with an inflow and an outflow as shown. The blood conduits may each include a scaffold coupled to a membrane. The central region between the distal pump portion **480** and the distal pump portion **490** is open and does not include a blood conduit, which allows blood pumped by the distal pump portion **480** to perfuse the renal arteries, as shown by the outflow of the distal pump portion **480**. The proximal pump portion **490** pumps blood towards the lower region of the body, as shown. The distal pump portion **480** is shown positioned just upstream to the renal arteries **452**, but the proximal end of the distal blood conduit may also be substantially aligned with one or more renal arteries so the outflow aids to perfuse one or both renal arteries. Exemplary anchors **420**, **420′**, **432** and **434** are also shown, which may be sized and configured to expand and contact the descending aorta **450** to anchor the pump portion in the ascending aorta.

(28) The pump portion in FIG. **4** may alternatively or in addition to incorporate any suitable feature described with respect to the embodiments in FIGS. **1-3**. For example, a variation of the pump shown in FIG. **4** includes one or more inflatable anchors, such as those shown in FIG. **2**, to help anchor the pump portion in the descending aorta. Alternatively, the embodiment in FIG. **4** may be sized such that outer surfaces (e.g., cylindrical surfaces) of one or both of pump portions **480** and **490** expand into direct contact with descending aorta tissue **450**, such as is the case in the embodiment shown in FIG. **3**. This is an example of suitable features from any of the embodiments herein being combined with suitable features of any of the other embodiments herein.

(29) In a variation on the blood pump shown in FIG. **4**, a proximal portion of the distal impeller extends to some extent proximally beyond the proximal end of the distal blood conduit, as is shown with the relative positions of the proximal impeller in FIGS. **1-3**. This may cause the outflow from the distal pump portion **480** to have more of a radial component than distal pump portions in which the distal impeller does not extend proximally beyond the proximal end of the blood conduit. This may help perfuse the renal arteries to a greater extent by directing more of the outflow from the distal pump portion radially outwards towards the ostia of the renal arteries.

(30) In a variation of the blood pump of FIG. **4**, the proximal pump portion **490** may include an impeller that is adapted and configured such that when the impeller is rotated, the impeller causes the proximal pump portion **490** to pump blood in a retrograde direction opposite that of the normal flow of blood, which is upward or in a superior direction in FIG. **4**. In this variation, the proximal pump portion **490**, when activated, pumps blood in a retrograde direction towards the ostia of the

renal arteries rather than pumping blood downstream. In these variations, the impellers may be driven by a common drive mechanism **472** (e.g., a drive cable and/or drive shaft). In some methods of use, the operation of the impellers of this variation may be controlled in a manner related to the cardiac cycle of the heart. In some uses, for example, the impellers may be activated synchronously with the cardiac cycle. For example without limitation, the distal and proximal impellers can be turned on or activated during systole or a portion of systole (e.g., peak systole), and deactivated during diastole or a portion of diastole. Controlling pump operation in this exemplary manner may help perfuse the renal arteries during systole.

(31) FIG. **4** also illustrates optional occluding member **492**. Occluding member **492** may be controllably inflated and deflated to controllably occlude blood flow through the descending aorta in the vicinity of the pump portion in coordination with the cardiac cycle to help perfuse the kidneys. With respect to the original description of FIG. **4** in which both impellers pump blood in an antegrade direction when rotated, the blood pump may include an optional inflatable and deflatable occluding member **492**. The occluding member may be coupled to sheath **430** as shown, but in alternative embodiments it may be secured to the catheter shaft. The inflatable and deflatable member **492** may be coupled to an outer surface of the sheath or the catheter shaft, and may be in fluid communication with a fluid source external to the patient via an inflation pathway extending through the catheter. In some embodiments, the occluding member **492** may be controllably inflated to occlude blood flow downstream the occluding member during systole to help perfuse the kidneys, and may be controllably deflated during diastole to allow blood to flow downstream.

(32) An aspect of this disclosure is related to intravascular blood pumps with pump portions that are sized and configured for placement in the vicinity of the junction of an inferior vena cava ("IVC") and one or more renal veins, and operated to reduce renal venous pressure. For example, in variations on FIGS. **1-4**, the blood pump may instead be positioned on the venous side of the vasculature so the pump portion is delivered and deployed in an IVC. FIG. **1-4** may be modified such that the renal arteries are labeled as renal veins, the descending aorta is an IVC, and the natural blood would flow out of the renal veins, into the IVC, and in a superior direction back towards the heart.

(33) FIG. **5** illustrates an exemplary pump portion **500** of an exemplary blood pump that may be include any suitable aspect of any of the blood pump described herein, and which may include any suitable structural component of any blood pump herein, including any of those shown in FIG. **1-4**. For example, pump portion **500** may be sized to expand and contact at least a portion of an IVC wall **513**. Additionally, the pump portion in the embodiment of FIG. **5** may include one or more anchoring/stabilizing members adapted to expand radially outward from the blood conduit and engage an IVC wall. FIG. **5** shows the pump portion **500** positioned in an IVC **513**, with the distal impeller **504** positioned superior to the renal veins **511**, and a proximal impeller **506** positioned inferior to the renal veins **511**. Relative positions of kidneys is also labeled.

(34) The impellers **504** and **506** are configured to pump blood from the pump inflow, through the expanded blood conduit or shroud **521**, and toward the pump outflow, as shown. One or more of the drive shaft rotation direction or impeller configuration causes the blood to be pumped in this direction through the expandable blood conduit **521**.

(35) In this exemplary embodiment, as shown in FIG. **5**, the blood pump includes one or more flow controllers **519** that are positioned, sized and configured to facilitate the flow of blood from outside the blood conduit and into the blood conduit. The pump portion **500** is expanded in a position such that the one or more flow controllers **519** (shown generally as circular or oval in this embodiment) are positioned generally at the location of the renal veins, such that blood flow out of the veins may enter into the blood pump fluid conduit through the one or more flow controllers. For example without limitation, any of the one or more flow controllers may comprise an opening, or in some embodiments any of the flow controllers herein may include a one-way valve (e.g., check valves) that allow blood to pass into the blood conduit but not out of the blood conduit. Other types of

suitable flow controllers may of course be incorporated into the pump portions herein. One or more flow controllers may be axially spaced (along the length of the pump portion) from other flow controllers. Flow controllers may be disposed circumferentially around a periphery of the blood conduit at any distance apart. For example, one or more flow controllers may be spaced 30 degrees, 60 degrees or 90 degrees apart. In any of the blood pumps herein, one or more flow controllers may not be circumferentially or axially spaced at regular intervals. In any of these embodiments, there may be from one—twenty separate flow controllers, such as from two to ten. FIG. **5** also shows exemplary delivery sheath **530**.
(36) In an alternative to that shown in FIG. **5**, a pump portion with multiple impellers may be expanded in place at a location that is superior to both veins, similar to the superior placement of the pump portion relative to the renal arteries shown in FIG. **1**. In these alternatives, the blood conduit would be downstream the renal veins, as opposed to only a portion of the pump portion being downstream as shown in FIG. **5**.
(37) With respect to any of the methods herein that position a pump portion in the arterial vasculature, such as in the embodiments in FIGS. **1-4**, the blood pump may be percutaneously and trans-luminally delivered to a portion of the descending aorta or renal artery of the patient via a femoral artery of the patient, for example, although other known entry locations and access pathways may also be used to deliver the pump portion to the target anatomy.
(38) With respect to any of the methods herein that position a pump portion in the venous vasculature, such as in the embodiments in FIG. **5**, the blood pump may be percutaneously and trans-liminally delivered to a portion of the or renal vein of the patient via a femoral vein, for example, although other known entry locations and access pathways may also be used to deliver the pump portion to the target anatomy, such as via a subclavian vein or jugular vein, for example. It is understood that the terms distal and proximal are used herein as is common to refer to the relative directions based on the pathway in which the blood pump is advanced, examples of which are shown in FIGS. **1-5**.
(39) In alternative embodiments, any of the pump portions herein may be positioned in a renal artery rather than in a descending aorta. For example, the pump portion may be delivered via a femoral artery and then advanced into a renal artery, optionally over a guidewire that has been previously advanced into a right or left renal artery. Similarly, any of the pump portions herein may be positioned in a renal vein rather than in an IVC. For example, the pump portion may be delivered via, a femoral vein and then advanced into a renal vein, optionally over guidewire that has been advanced into a right or left renal vein. In any of these embodiments, pump portions from separate blood pumps can be positioned bilaterally in first and second renal arteries, or in first and second renal veins, or in any combination of renal arteries and renal veins (e.g., up to four pump portions delivered through separate sheaths). In any of these alternative embodiments, a pump portion positioned in a renal vein or a renal artery may optionally be a single impeller pump rather than a multiple impeller pump.

Claims

1. A method of supporting circulation in a descending aorta of a subject, comprising: advancing an expandable pump portion of a blood pump into a descending aorta of a subject, the pump portion comprising an expandable blood conduit comprising a scaffold coupled to a membrane between a pump inflow and a pump outflow, the pump portion further including first and second expandable impellers at least partially within the blood conduit; expanding the expandable blood conduit to an expanded and deployed configuration within the descending aorta; expanding the first expandable impeller into an expanded configuration at least partially within the blood conduit; expanding the second expandable impeller into an expanded configuration at least partially within the blood conduit and having a portion that extends beyond the expandable blood conduit; and rotating the

first and second expandable impellers to thereby move blood into the pump portion, through the blood conduit, and out of the pump portion, wherein rotation of the portion of the second expandable impeller that extends beyond the expandable blood conduit causes the pump outflow to include a radial flow component.

2. The method of claim 1, wherein the rotating step moves blood through the blood conduit at a rate of at least 3.5 L/min.

3. The method of claim 1, wherein moving blood out of the pump outflow perfuses at least one renal artery.

4. The method of claim 1, wherein expanding the expandable blood conduit comprises expanding the expandable blood conduit to a deployed configuration within the descending aorta such that the pump outflow is upstream or aligned with a renal artery, and wherein the outflow is at least partially directed radially into the renal artery due to the position of the blood conduit in the descending aorta.

5. The method of claim 1, wherein expanding the expandable blood conduit to a deployed configuration within the descending aorta comprises expanding the expandable blood conduit to a deployed configuration near a renal artery such that the pump outflow perfuses at least one renal artery.

6. The method of claim 1, wherein expanding the expandable blood conduit within the descending aorta comprises expanding a distal scaffold to an expandable configuration that provides radial support to the blood conduit at the location of the first impeller.

7. The method of claim 6, wherein expanding the expandable blood conduit within the descending aorta comprises expanding a proximal scaffold to an expandable configuration that provides radial support to the blood conduit at the location of the second impeller.

8. The method of claim 1, wherein expanding the expandable blood conduit comprises expanding a central region of the blood conduit that is between the first and second impellers, wherein the central region is more flexible than regions of the blood conduit that surround the first and second impellers.

9. The method of claim 1, wherein expanding the expandable blood conduit to a deployed configuration within the descending aorta comprises expanding the expandable blood conduit to a deployed configuration within the descending thoracic aorta.

10. A method of supporting circulation in a descending aorta of a subject, comprising: advancing an expandable pump portion of a blood pump into a descending aorta of a subject, the pump portion including a distal impeller and a proximal impeller; expanding a distal expandable blood conduit comprising a scaffold coupled to a membrane into an expanded configuration aligned with or upstream to a renal artery; expanding the distal impeller to an expanded configuration at least partially within the distal expandable blood conduit and having a portion that extends beyond the distal expandable blood conduit; expanding a proximal expandable blood conduit into an expanded configuration downstream from the renal artery; expanding the proximal impeller to an expanded configuration at least partially within the proximal expandable blood conduit; rotating the distal impeller to move blood into a distal end of the distal expandable blood conduit, through the distal expandable blood conduit, and out of a proximal end of the distal expandable blood conduit, the portion of the distal impeller extending beyond the distal expandable blood conduit causing blood moving out of the distal expandable blood conduit to include a radial flow component; and rotating the proximal impeller to move blood through the proximal expandable blood conduit.

11. The method of claim 10, wherein rotating the distal impeller causes blood to move out of the proximal end of the distal expandable blood conduit and perfuse the renal artery, optionally also perfusing a second renal artery.

12. The method of claim 10, wherein rotating the distal and proximal impellers moves blood past the distal and proximal impellers in an antegrade direction.

13. The method of claim 10, wherein rotating the distal impeller moves blood past the distal

impeller in an antegrade direction, and wherein rotating the proximal impeller moves blood past the proximal impeller in a retrograde direction toward the renal artery.

14. The method of claim 13, wherein rotating the distal and proximal impellers is performed discontinuously and in a manner that is related to one or more aspects of the subject's cardiac cycle.

15. The method of claim 14, wherein rotating the distal and proximal impellers occurs during at least a portion of systole, and wherein the distal and proximal impellers are not rotated during at least a portion of diastole, optionally during any of diastole.

16. The method of claim 10, wherein rotating the proximal impeller moves blood into a distal end of the proximal expandable blood conduit, through the proximal expandable blood conduit, and out of a proximal end of the proximal expandable blood conduit.

17. The method of claim 10, wherein expanding a distal expandable blood conduit into an expanded configuration aligned with or upstream to a renal artery comprises expanding a distal expandable scaffold.

18. The method of claim 10, wherein expanding a proximal expandable blood conduit comprises expanding a proximal expandable scaffold.
